

Foundation Practice Book

Guided retrieval -> fresh cumulative attempt -> validation -> explain-back

QUESTION

MEASURE

GRAIN

TRUST

EVIDENCE

How to use this practice book

This book contains tasks but no full solutions. Short answers are acceptable; the standard is coherent reasoning. Use the lesson book to learn first. Then attempt from memory. Save the attempt before opening protected review.

Attempt sequence

- 1 Do pages 2-10 as lesson retrieval checks.
- 2 Do the guided cumulative case with the hint ladder visible.
- 3 Close the hints. Complete the fresh cumulative case independently.
- 4 Use the validation page to check your own work.
- 5 Explain the case aloud or in writing.
- 6 Save evidence. Only then may the protected Practice Review open if needed.

HONESTY RULE

Do not convert review material into notes before the attempt. Review exists to diagnose a real attempt, not to become a hidden answer sheet.

O1 - How This Course Works

Complete from your own reasoning. If unsure, write the uncertainty instead of inventing a rule.

1	Explain Completed vs Mastered in your own words.	
2	Why does a changed independent task exist?	
3	If one competency is weak, what happens next?	
4	Why can estimated hours never create a pass?	

Evidence: save this page or equivalent notes. Mark uncertainty before opening any protected review.

O2 - Files, Folders and Safe Data Handling

Complete from your own reasoning. If unsure, write the uncertainty instead of inventing a rule.

1	Write the Analyst_Lab folder tree from memory.	
2	Where does an untouched source belong?	
3	Where does a transformed working copy belong?	
4	Explain Copy vs Move and why the only raw source is never intentionally edited.	

Evidence: save this page or equivalent notes. Mark uncertainty before opening any protected review.

O3 - Using the Learning System

Complete from your own reasoning. If unsure, write the uncertainty instead of inventing a rule.

1	You return tomorrow and forgot where to resume. Reconstruct the route.	
2	Which review artifact stays closed before a saved attempt?	
3	Which Gate artifact stays closed unless explicitly assigned?	
4	When is a changed fresh retry required?	

Evidence: save this page or equivalent notes. Mark uncertainty before opening any protected review.

F1 - What Does a Data Analyst Actually Do?

Complete from your own reasoning. If unsure, write the uncertainty instead of inventing a rule.

1	Map "Sales seem worse" through the eight analyst stages.	
2	Name one tool-first failure mode.	
3	Why does validation come before communication?	
4	Why does an analyst support the decision rather than automatically own the business decision?	

Evidence: save this page or equivalent notes. Mark uncertainty before opening any protected review.

F2 - From Vague Request to Analytical Question

Complete from your own reasoning. If unsure, write the uncertainty instead of inventing a rule.

1	Request: "Complaints are too high lately." Identify stakeholder and decision.	
2	Write one clarification question for Decision, Comparison, Scope, Time and Meaning.	
3	Separate two supplied/confirmed facts from two provisional assumptions.	
4	Write one bounded analytical question that does not assume a cause.	

Evidence: save this page or equivalent notes. Mark uncertainty before opening any protected review.

F3 - Metrics and KPIs

Complete from your own reasoning. If unsure, write the uncertainty instead of inventing a rule.

1	Define On-Time Delivery Rate in plain English.	
2	State numerator and eligible denominator.	
3	State unit, time basis, includes/excludes and definition owner/approver.	
4	Give one boundary case and explain why a target is not the metric itself.	

Evidence: save this page or equivalent notes. Mark uncertainty before opening any protected review.

F4 - Grain and Required Data

Complete from your own reasoning. If unsure, write the uncertainty instead of inventing a rule.

1	Table A is one row per order. Table B is one row per order-product line. State both grains.	
2	Which identifier should be unique at each grain?	
3	Explain when repeated OrderID is legitimate.	
4	For a region-month sales summary, state intended result grain and minimum fields/business rules needed.	

Evidence: save this page or equivalent notes. Mark uncertainty before opening any protected review.

F5 - Data Quality

Complete from your own reasoning. If unsure, write the uncertainty instead of inventing a rule.

1	For a service-visit table, write one scenario-specific check for completeness, uniqueness, validity, consistency and timeliness.	
2	For each failed check, state why the business answer might be harmed.	
3	Name one value that could look unusual without automatically being wrong.	
4	Explain why missing/duplicate-looking data is an investigation signal rather than an auto-delete instruction.	

Evidence: save this page or equivalent notes. Mark uncertainty before opening any protected review.

F6 - Analysis Validation and the Complete Analyst Brief

Complete from your own reasoning. If unsure, write the uncertainty instead of inventing a rule.

1	For a subscription-cancellation concern, write the four Analyst Brief section names from memory.	
2	State one internal validation check and one reconciliation idea.	
3	State one assumption and one limitation separately.	
4	Write a testable Definition of Done and a Ready/Not Ready judgment.	

Evidence: save this page or equivalent notes. Mark uncertainty before opening any protected review.

Guided cumulative case - FreshNest Home Services

Business concern: "Customers say technicians arrive late. Tell us which service areas need attention first."

Schema card - service_visits: one intended row per VisitID with VisitID, CustomerID, ServiceArea, ScheduledStart, ArrivalTime, Status, TechnicianTeam, LastUpdatedAt.

Schema card - visit_events: many rows per VisitID with EventID, VisitID, EventTime, EventType.

Known ambiguity: no approved grace minutes; cancelled/rescheduled visits may or may not be eligible; there is no trustworthy field proving traffic or staffing as a cause.

Follow the structure once

- 1 PROBLEM: identify likely decision owner, decision and one bounded non-causal question.
- 2 MEASURE + DATA: propose an arrival metric but mark the grace/eligibility rule as needing approval; state both source grains and intended result grain.
- 3 TRUST CHECKS: write one check for each of the five quality dimensions; add validation, reconciliation and a sanity check.
- 4 DONE + DEFENSE: write testable acceptance criteria, assumptions/limitations and Ready/Not Ready.

HINT LADDER

1) Start with the decision. 2) For a rate, ask "out of which eligible visits?" 3) Repeated VisitID in visit_events can be expected. 4) Do not claim traffic/staffing caused lateness without evidence.

Fresh independent cumulative case - BrightBasket Grocery

Business concern: "Customers seem to be waiting longer for order pickup. Find the stores and time periods that need attention first."

Schema card - pickup_orders: one intended row per OrderID with OrderID, StoreID, CustomerID, ReadyAt, PickupAt, Status, OrderChannel, LastUpdatedAt.

Schema card - order_events: many rows per OrderID with EventID, OrderID, EventTime, EventType.

Official reference: Store Operations publishes completed pickup counts by store/week, but not wait-time metrics. No approved wait-time target is supplied.

Known ambiguity: "wait" could mean ReadyAt->PickupAt or another business definition; abandoned/cancelled orders may have different eligibility; event history does not prove root cause.

Build the four-section Analyst Brief

1 PROBLEM	Stakeholder/user; decision; clarification questions; bounded analytical question; confirmed facts vs assumptions.
2 MEASURE + DATA	Metric/KPI rationale; calculation rule; owner/approver; source grain; result grain; minimum required data.
3 TRUST CHECKS	Completeness; uniqueness; validity; consistency; timeliness; analysis validation; reconciliation; sanity check.
4 DONE + DEFENSE	Testable Definition of Done; assumptions/limitations; Ready/Not Ready; defend denominator/grain/one unsupported conclusion/trust evidence.
PROTECTED BOUNDARY	Do not open the Practice Review while this case is in progress. Save the attempt first, even if it is incomplete.

Validate your cumulative attempt

- Your question is bounded and does not assume a cause.
- Any rate has an explicit numerator and eligible denominator or clearly marks the business rule as unresolved.
- Source grain and intended result grain are stated. Expected repeated IDs are not called duplicates automatically.
- Five quality checks name actual fields/rules from the scenario.
- Validation and reconciliation compare like with like or explain why a supplied reference is not directly comparable.
- Assumptions and limitations are visible.
- Definition of Done is testable.
- Ready/Not Ready is reasoned, not guessed.

Teach it back

Without reading your brief, explain the full reasoning in 90-120 seconds: decision -> question -> metric -> grain/data -> five quality checks -> validation/reconciliation -> Definition of Done -> one limitation.

Save evidence

Record the saved file/location and your uncertain competency in the evidence log. If all checks pass, close this book and wait for Gate A assignment. If one or more checks are uncertain, save the attempt first and then open the protected Practice Review.